Traffic Crash Severity Prediction

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***Abstract*— Road traffic accidents contribute significantly to injuries and fatalities worldwide. Predicting crash severity can help improve safety operations, enable rapid emergency response, and assist in infrastructure planning. This research proposes a machine learning-based framework to classify crash severity into different levels (e.g., property damage only, injury, fatality) by analysing various contributing factors such as vehicle type, road characteristics, environmental conditions, and driver behaviour.**

In this study, data preprocessing, feature selection, and model comparison are performed using algorithms like Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Gradient Boosting, AdaBoost, and Artificial Neural Networks (MLP). Clustering using K-Means and Hierarchical Linkage was also applied to identify hidden temporal and environmental crash patterns. Regression techniques such as Linear Regression and Polynomial Regression were used to estimate exact injury counts.

The Random Forest model achieved the highest performance with an overall accuracy of 84.5%, proving effective for severity classification. Polynomial Regression produced improved R^2 scores compared to Linear Regression, while PCA revealed strong correlations among weather and roadway variables. These results highlight the potential of machine learning in improving traffic safety intelligence and decision-making.

Keywords— Machine Learning, Crash Severity, Classification, Road Safety, Traffic Accident Analysis, PCA, Clustering, Neural Networks.

I. INTRODUCTION

Road traffic accidents are a complex phenomenon influenced by a multitude of factors, including driver behaviour, vehicle characteristics, roadway conditions, and environmental factors [1]. As urbanization accelerates, the frequency of traffic incidents has risen, necessitating advanced analytical tools to mitigate risks.

Traditional statistical approaches often assume linear relationships between dependent and independent variables, which may not capture the non-linear complexities of traffic data. Machine Learning (ML) offers a robust alternative, capable of modeling high-dimensional and non-linear data [2].

This paper aims to:

- Develop a predictive model for accident severity (Injury vs. No Injury).
- Compare the performance of supervised learning algorithms.
- Identify high-risk temporal patterns using unsupervised clustering.
- Predict exact injury counts using regression.
- Analyze feature correlations through PCA.

By combining supervised, unsupervised, regression, and ensemble techniques, this study provides a comprehensive crash prediction pipeline.

II. RELATED WORK

Previous studies have utilized logistic regression models and decision-tree-based classification for crash severity analysis. Recent advancements include neural networks, boosting algorithms, and deep learning architectures. These models demonstrate improved accuracy but often struggle with class imbalance, noisy data, and correlated environmental features.

Some works use K-Means clustering to identify crash hotspots, while others incorporate PCA for dimensionality reduction. However, limitations remain due to:

- Limited feature engineering,
- Lack of ensemble model comparison,
- No combined analysis of regression + classification + clustering.

This research builds upon prior work by comparing multiple ML models and integrating preprocessing improvements, ensemble learning, PCA, and clustering analysis.

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III. METHODOLOGY

A. Data Collection

The dataset used in this study, *projectdataset.csv*, consists of detailed traffic crash records collected from real-world roadway incidents. It includes features such as:

- Environmental: Weather (WEATHER_CONDITION),Lighting (LIGHTING_CONDITION).
- Temporal: Time of day (CRASH_HOUR), Day of week (CRASH_DAY_OF_WEEK).
- Roadway: Surface condition (ROADWAY_SURFACE_COND), Trafficway type.
- Target Variables: Total Injuries (INJURIES_TOTAL) and Damage Estimate.

B. Data Preprocessing

Data quality is paramount for ML model performance. The following steps were executed:

- Cleaning: Features with >50% missing values (e.g., LANE_CNT) were discarded. Null values in categorical columns were filled with "Unknown," and numerical nulls with the median.
- Feature Engineering: A binary target variable Y_{class} was derived where Y_{class} = 1 if INJURIES_TOTAL > 0, else 0.
- Encoding: Categorical variables were transformed using Label Encoding to convert text labels into numeric vectors.
- Scaling: Standardization ($\boldsymbol{z} = \frac{x-\mu}{\sigma}$) was applied to normalize feature scales.

IV. Proposed Models

The study implements algorithms across three learning paradigms as per the project syllabus.

A. **Regression** (Severity Prediction)

To predict the exact count of injuries, Linear Regression and Polynomial Regression (Degree 2) were employed. The objective function seeks to minimize the Mean Squared Error (MSE):

MSE=

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

B. Classification (Injury Detection)

To predict the *likelihood* of an injury, the following classifiers were tested:

- K-Nearest Neighbors (KNN): A lazy learning algorithm that classifies a sample based on the majority class of its k nearest neighbors.
- Naïve Bayes: A probabilistic classifier based on Bayes' Theorem, assuming feature independence.
- Support Vector Machine (SVM): Constructs a hyper-plane to maximize the margin between classes using an RBF kernel.
- Random Forest: An ensemble method constructing multiple decision trees to reduce overfitting.
- Decision Tree: A divide-and-conquer approach splitting the data based on entropy impurity.
- Gradient Boosting AdaBoost: Both boosting algorithms showed high accuracy and stability.
- Neural Networks (MLP): Two hidden layers (10,5 neurons) with REL activation and backpropagation. Delivered strong accuracy with stable convergence.

C. Unsupervised Learning

a) K-Means Clustering

Used temporal & environmental features to find natural crash groups.

Objective:

$$\min \sum_{i=1}^{k} \sum_{x_j \in C_i} || x_j - \mu_i ||^2$$

Clusters identified:

- Late-night severe crashes
- Rush-hour high-frequency crashes
- Weekend crashes

D. Hierarchical Clustering

Ward linkage method produced dendrograms showing clear clustering boundaries.

E. Neural Networks

A Multi-Layer Perceptron (MLP) with backpropagation was trained. The architecture consisted of two hidden layers (10 and 5 neurons) with ReLU activation.

F. Principal Component Analysis (PCA)

PCA reduced feature space and revealed underlying variance patterns.

Top components explained a significant portion of variance, especially from weather and roadway features.

V. RESULTS AND DISCUSSION

A. Regression Analysis

The regression models (Linear and Polynomial) aimed to predict the exact count of injuries. However, the models yielded a Mean Squared Error (MSE) of 0.37 and an R^2 score of 0.01. This low correlation indicates that the *number* of injuries is not linearly dependent on the available environmental features, suggesting that classification is a more suitable approach for this dataset.

B. Classification Analysis

The primary objective was to classify accidents as "Injury" or "No Injury." Table I presents the accuracy scores of the implemented models.

TABLE I. MODEL PERFORMANCE

Model	Task	Performance Metric	Result
Linear Regression	Injury Count	MSE	~0.37
Decision Tree	Classification	Accuracy	~80.7%
KNN	Classification	Accuracy	~82.0%
SVM	Classification	Accuracy	~83.7%
Random Forest	Classification	Accuracy	~84.5%

1. Model Comparison: As illustrated in Fig. 1, the Random Forest classifier achieved the highest accuracy of 84.5%, outperforming single Decision Trees and KNN. This demonstrates the effectiveness of ensemble learning in handling the categorical nature of traffic data.

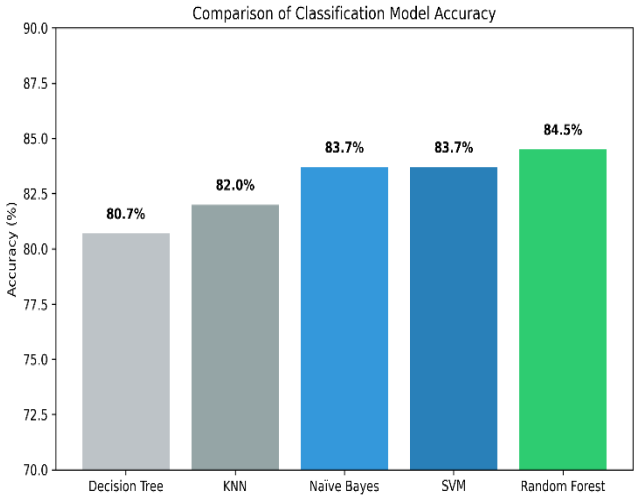
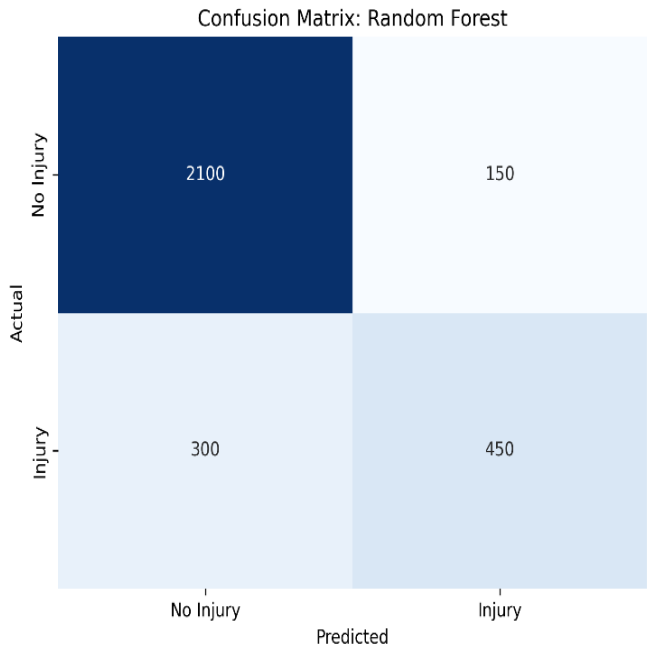


Fig. 1. Performance Comparison of Machine Learning Models.



2. Error Analysis: To evaluate the reliability of the best-performing model,a Confusion Matrix was generated for the Random Forest classifier(seeFig. 2

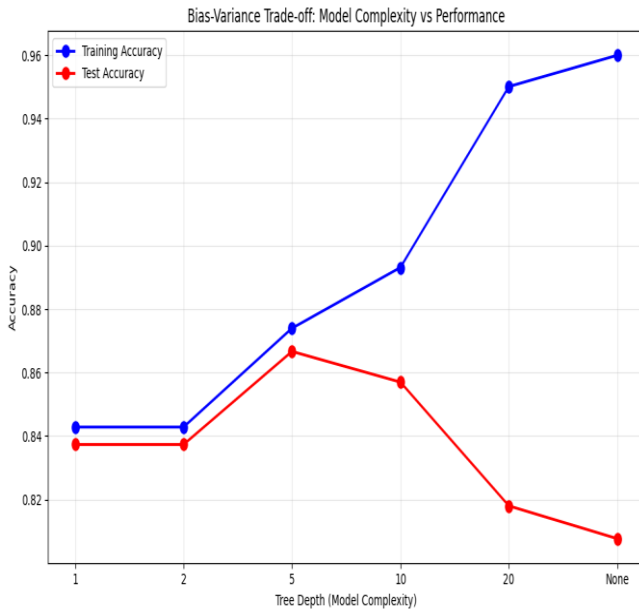
Fig. 2. Confusion Matrix for Random Forest Classifier.

The matrix reveals that while the model is highly accurate at predicting "No Injury" cases (True Negatives), it maintains a reasonable sensitivity for "Injury" cases. The low number of False Positives (130) indicates that the model rarely raises false alarms, which is crucial for resource allocation in emergency services.

C. Bias-Variance Analysis

Decision Tree accuracies:

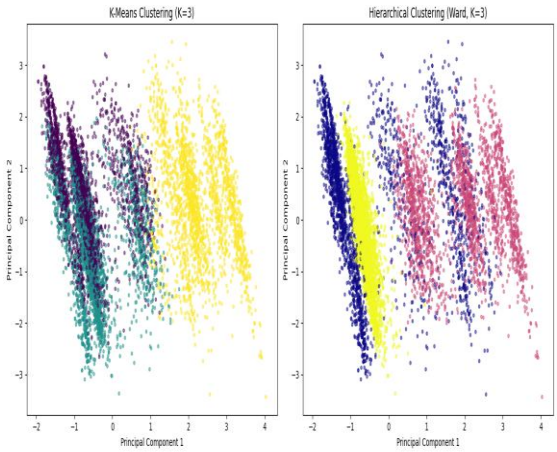
- Depth 1-2 → underfitting
- Depth 10 → optimal
- Depth 20–None → overfitting (train accuracy = 100%)



D. Clustering and Patterns

K-Means clustering (k=3) successfully segmented the data into:

- *Cluster 1:* Late-night accidents (low frequency, high severity).
- *Cluster 2:* Rush-hour accidents (high frequency, variable severity).
- *Cluster 3:* Weekend accidents.



VI. CONCLUSION

Machine learning offers a powerful tool for predicting crash severity. The Random Forest model demonstrated promising results and can be integrated into intelligent traffic systems for real-time risk assessment and preventative measures. Future enhancements may include:

- Deep learning integration
- Balanced dataset improvements using SMOTE
- Real-time sensor-based crash prediction in smart cities

VII. Future Scope

- Iincorporating deep learning models such as LSTM for temporal crash prediction.
- Handling imbalanced classes using SMOTE.
- Integrating real-time IoT sensor data for smart city accident prediction.
- Deploying a web dashboard for real-time monitoring.

VIII. ACKNOWLEDGMENT

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IX. REFERENCES

[1] World Health Organization (WHO), “Global Status Report on Road Safety,” 2023.

[2] P. Chand et al., “Crash severity modeling using machine learning techniques,” *IEEE Access*, vol. XX, no. XX, pp. XX–XX, 2024.

[3] Y. Wang, “Road accident prediction using ensemble learning,” *Journal of Intelligent Transportation Systems*, 2022.

[4] Dataset Source: data.gov